

Distinct effects of phyllosphere and rhizosphere microbes on invader *Ageratina adenophora* during its early life stages

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Abstract

Microbes strongly affect invasive plant growth. However, how the phyllosphere and rhizosphere soil microbes distinctively affect seedling mortality and the growth of invasive plants across ontogeny under varying soil nutrient levels remains unclear. In this study, we used the invader *Ageratina adenophora* to evaluate these effects in plant growth chambers. We found that leaf litter harboured more potential pathogens and thus had more adverse effects on seed germination and seedling survival than soil inoculation. Microbial inoculation at different growth stages altered the microbial community and microbial functions of seedlings, and earlier inoculation had a more adverse effect on seedling survival and growth. In most cases, the soil nutrient level did not affect microbe-mediated seedling growth and the relative abundance of the microbial community and functions involved in seedling growth. The effects of some microbial genera on seedling survival are distinct from those on growth. Moreover, the *A. adenophora* seedling-killing effects of fungal strains isolated from dead seedlings by nonsterile leaf inoculation litter exhibited significant phylogenetic signals, by which strains of *Allophoma* and *Alternaria* generally caused high seedling mortality. Our study stresses the essential role of *A. adenophora* litter microbes in population establishment by regulating seedling density and growth.

eLife assessment

This study investigates plant-microbe interactions for an invasive plant, *Ageratina adenophora*. The findings are **valuable** in advancing our understanding of how leaf and soil microbes separately affect its performance, with **solid** experimental evidence revealing the importance of litter microbes in shaping *A. adenophora* populations. The work will be of interest to invasion biologists.

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Introduction

Plant-modified soil properties affect the performance of plants, which are termed ‘plant–soil feedbacks (PSFs)’. PSFs can affect species coexistence (Bever et al., 1997 [↗](#); van der Putten et al., 2013 [↗](#)) and local plant community composition and dynamics (Bauer et al., 2017 [↗](#); Kardol et al., 2007 [↗](#); Teste et al., 2017 [↗](#)). For plant invasion, PSFs are usually positive because of escaping soil pathogens and recruiting some beneficial microbes (Liu et al., 2023 [↗](#); Mitchell & Power, 2003 [↗](#); Xu et al., 2012 [↗](#)) or negative effects because of accumulating local pathogens (Callaway et al., 2013 [↗](#); Flory & Clay, 2013 [↗](#); Zhang et al., 2020 [↗](#)).

Similar to soil, leaf litter can also affect plant growth, species diversity and community structure (Liu et al., 2017 [↗](#); Ma et al., 2020 [↗](#); Olson & Wallander, 2002 [↗](#)), thus playing important roles in population establishment and community dynamics (Jessen et al., 2023 [↗](#); Lamb, 2008 [↗](#); Xiong & Nilsson, 1999 [↗](#)). However, related research has focused mainly on physical (e.g., maintaining soil moisture and temperature, increasing nutrition and reducing light) or chemical effects (e.g., releasing allelochemicals) (Demey et al., 2013 [↗](#); Jessen et al., 2023 [↗](#); Möhler et al., 2018 [↗](#); Zhang et al., 2017 [↗](#)) but has rarely focused on leaf microbial effects. Until 2017, Whitaker et al. (2017) [↗](#) extended the PSF to aboveground tissues (including leaf, stem and floral tissues), termed “plant–phyllosphere feedbacks (PPFs)”, and found that all four Asteraceae species experienced stronger negative PPFs than PSFs. Subsequently, this team further verified that all ten tested Asteraceae plants experienced negative PPFs (Zaret et al., 2021 [↗](#)). The lack of strong mutualists and relatively high abundance of pathogens in the phyllosphere may account for the negative PPFs.

In addition to microbial sources (i.e., soil vs leaf litter), ontogeny (seedling growth stage) and soil nutrient levels can affect plant–microbe interactions. For example, seedlings showed distinct sensitivity during the growth stage, and younger seedlings were more susceptible to infection by soil microbes because of fewer defense resources (Geisen et al., 2021 [↗](#); Jevon et al., 2020 [↗](#)). Interestingly, leaf litter has an adverse effect on seedling emergence but a positive effect on later plant growth (Abbas et al., 2023 [↗](#); Möhler et al., 2021 [↗](#); Zhang et al., 2022 [↗](#)); litter also has a stronger negative effect on earlier vegetation growth than on the elder (Loydi et al., 2013 [↗](#); Wang et al., 2022 [↗](#); Xiong & Nilsson, 1999 [↗](#)). Moreover, plants enrich distinct microbes under different nutrient conditions and affect plant performance (Dostál, 2021 [↗](#); Gustafson & Casper, 2004 [↗](#); Widdig et al., 2020 [↗](#)). For example, the bacterial diversity in duckweed plants was reduced under nutrient–deficient conditions, but the abundance of Firmicutes increased (Bunyoo et al., 2022 [↗](#)), and members of Firmicutes have been reported to promote plant stress tolerance (Xu et al., 2018 [↗](#)). Ai et al. (2018) [↗](#) reported that nutrient additions cause crops to enrich some bacteria and fungi from soil and increase yield.

Ageratina adenophora (Sprengel) R.M. King & H. Robinson (Asteraceae), known as Crofton weed or Mexican devil weed, has invaded more than 30 countries and areas, including South Africa, Australia, New Zealand, Hawaii, India and China (Gu et al., 2021 [↗](#); Julien & Griffiths, 1998 [↗](#)). It is a perennial weed and can produce high yields of seeds with a high germination rate (Lu et al., 2006 [↗](#); Parsons WT, 1992 [↗](#)). This weed commonly grows in monocultures, but a high density of seedlings is not common in the wild. Previous studies have shown that *A. adenophora* can enrich the beneficial soil microbial community to facilitate invasion (Niu et al., 2007 [↗](#); Zhao et al., 2019 [↗](#)); in contrast, *A. adenophora* leaves harbor diverse fungal pathogens that can cause adverse effects on itself seed germination and growth (Kai Fang et al., 2019 [↗](#); K. Fang et al., 2021 [↗](#)). Thus, it is interesting to determine whether leaf microbes play a distinct role from soil microbes in regulating *A. adenophora* seedling density and whether these effects change with the *A. adenophora* growth stage and soil nutrition level.

In this study, we inoculated *A. adenophora* with soil or leaf litter at three stages, 0 d, 21 d, and 28 d after sowing, and transplanted seedlings to grow in soils with high or low nutritional levels. We first determined the germination, seedling survival and growth of the *A. adenophora* plants. Then, we characterized the bacterial and fungal communities of the soils and leaf litter as inoculation sources, as well as the microbial communities enriched in the leaves and roots of the *A. adenophora* seedlings after growing; we also isolated the fungi associated with the dead seedlings and tested their seedling-killing effects on *A. adenophora*. Finally, we correlated the microbial community with *A. adenophora* seedling mortality and growth.

We hypothesized that the microbial communities associated with leaf litter and rhizosphere soils can account for the differential effects on *A. adenophora* seedling mortality and growth during different growth stages when growing under different nutrient conditions. We expected that 1) leaf litter would have more adverse effects on seed germination, seedling survival and growth than soil, as leaf litter often harbours more plant pathogens, and 2) inoculation at different growth stages would change the microbial community enriched by seedlings and thus affect seedling growth. Earlier inoculation had a greater adverse impact on seedling growth than later inoculation, as younger seedlings were more sensitive to pathogen infection than older seedlings. 3) The nutrient level influences seedlings to recruit microbes and thus affects seedling growth.

Results

Effects of leaf litter and rhizosphere soil on the mortality and growth of *A. adenophora* seedlings

At the G0 timepoint, sterile leaf inoculation significantly delayed germination time more than did soil and sterile leaf inoculation, as well as the control (nothing inoculated) (**Fig. 1a**, $P < 0.05$). Leaf and soil inoculation had no distinct effects on the germination rate (**Fig. 1b**, $P > 0.05$). In addition, the inoculation of sterile and non-sterile leaves at G0 caused a high mortality rate (19.7% vs 96.7%) for seedlings growing in petri dishes (**Fig. 1c**, Fig. S1). Only nonsterile leaves caused a low percentage of seedling death (8.4%) when the seedlings were inoculated at G21 (**Fig. 1c**). Two weeks after transplanting these seedlings into the cups, leaf inoculation caused significantly greater seedling mortality than did soil inoculation ($P < 0.001$); the nonsterile sample caused greater seedling mortality than did the sterile sample, especially leaf inoculation during the G0 and G21 periods. Moreover, nonsterile leaf inoculation at earlier stages significantly increased seedling mortality compared with that at later stages (**Fig. 1d**, $P < 0.05$). However, seedling mortality did not differ between the high- and low-nutrient conditions, regardless of leaf or soil inoculation (**Fig. 1d**, both $P > 0.05$).

With the exception of nutrient level, inoculation source and time, as well as their interaction with nutrient level, significantly affected the microbial role in the total dry biomass of *A. adenophora* ($P < 0.05$, **Fig. 2a**). Soil and leaf microbial effects on seedling biomass interacted with soil nutrient level and inoculation time: when inoculated at the G0 timepoint, both soil and leaf microbes had adverse effects on *A. adenophora* growth, as all seedlings inoculated with nonsterile leaves died, and those inoculated with nonsterile soil grew poorer than those inoculated with sterile soil under both low- and high-nutrition conditions; when inoculated at the G21 timepoint, both soil and leaf microbes had significantly positive effects only under high-nutrition conditions; when inoculated at the G28 timepoint, only soil microbes promoted seedling growth under high-nutrition conditions; and when inoculated at the G21+28 timepoint, only leaf microbes had a significantly positive effect on seedling growth under low-nutrition conditions (**Fig. 2b**).

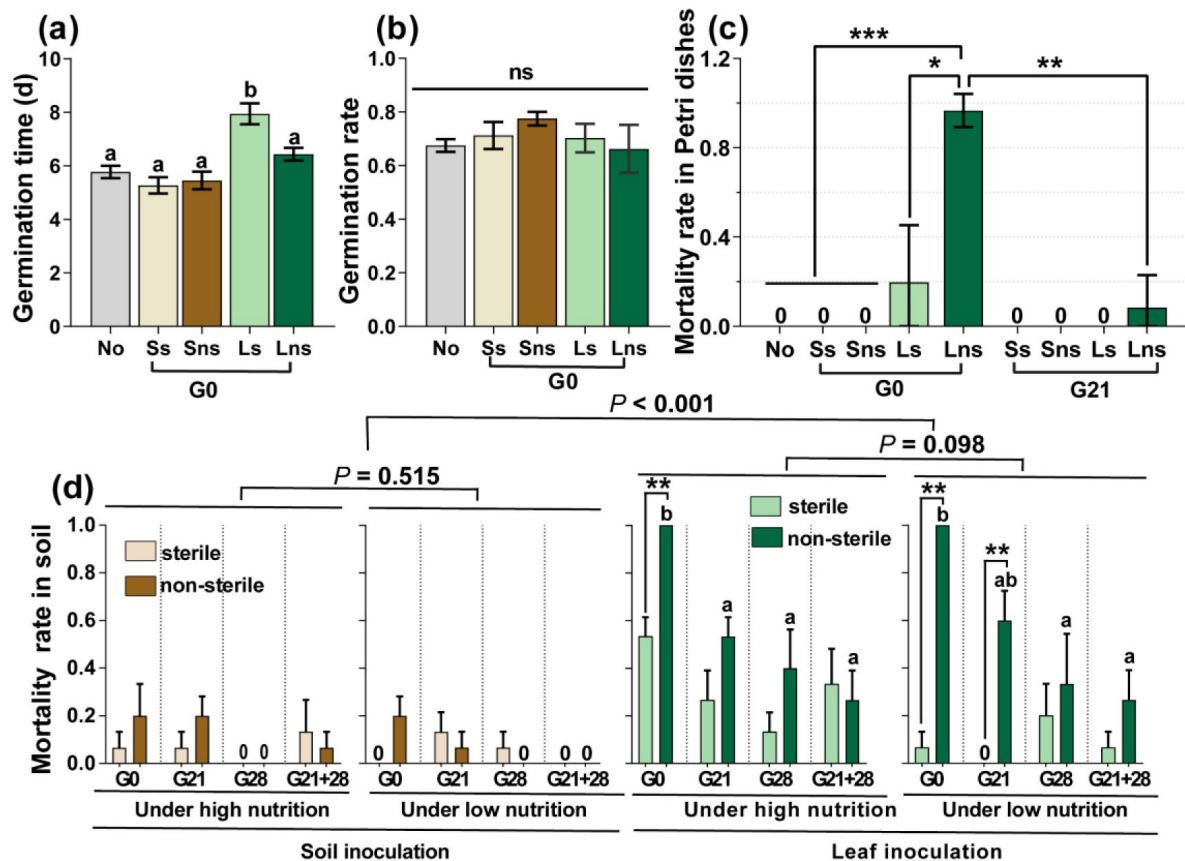


Figure 1.

A. adenophora seed germination and seedling mortality

Seed germination time and rate after soil or leaf inoculation at the G0 (a-b) and seedling mortality rate after soil or leaf inoculation at the G0 or G21 and subsequent growth in Petri dishes (c). Seedling mortality rate after two weeks when seedlings were transplanted into soil (d). No: nothing inoculated, Ss: sterile soil; Sns: non-sterile soil; Ls: sterile leaf; Lns: non-sterile leaf; G0, inoculated on the day of germination; G21, inoculated on the 21st day after germination; G28, inoculated on the 28th day after germination; G21+28, inoculated on both the 21st and 28th days after germination. * $P < 0.05$, *** $P < 0.001$. Error bars depict the standard error. Different lowercase letters represent significant differences among the different inoculation time treatments ($P < 0.05$). No lowercase letters indicate no differences among the four inoculation time treatments under the same nutrient level ($P > 0.05$).

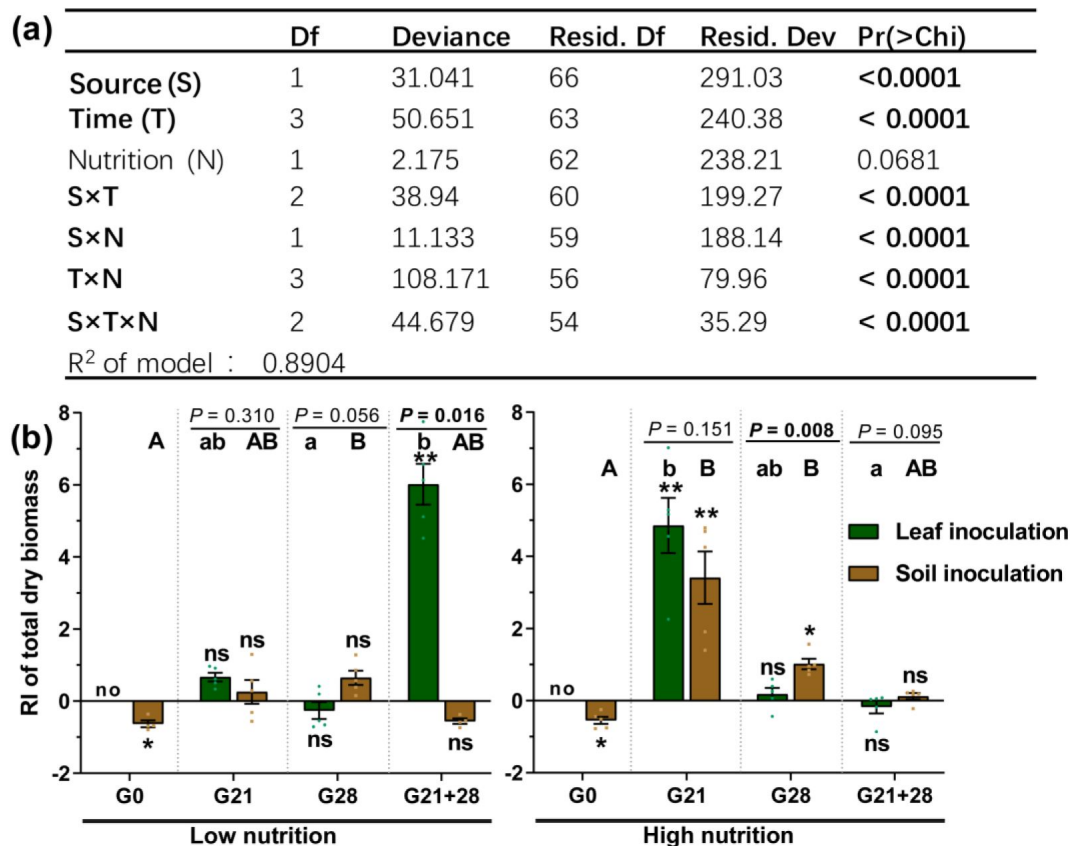


Figure 2.

GLM analysis (a) and comparison (b) of the role of microbes in seedling growth based on total dry biomass (RI) between leaf and soil inoculated at low and high nutrient levels across four inoculation time points.

Source refers to the leaf or soil inoculation source, time refers to the four inoculation time treatments, and nutrition refers to the nutrient level of the soil. For G0, G21, G28 and G21+28, see [Figure 1](#). Error bars depict the standard error. The difference in the RI among the different inoculation time treatments is indicated by different capital and lowercase letters for the soil and leaf inoculation treatments, respectively. The asterisks indicate that the RIs are significantly different from zero. "No" indicates no surviving seedlings at harvest. $P < 0.05$ is shown in bold.

Correlations of the microbial community and potential functions of inocula with *A. adenophora* seedling mortality at the early stage

The soil and leaf inocula had distinct microbial diversity, community compositions and potential functions. The microbial diversity and richness were greater in the soil than in the leaf litter (Fig. S2). Top three core bacteria were *Rhodoplanes* (5.65%), *Bradyrhizobium* (4.80%) and some unclassified Alphaproteobacteria (4.56%) for soil and *Pseudomonas* (30.33%), *Massilia* (17.45%) and *Sphingomonas* (16.35%) for leaf (Fig. 3ab). Top three core fungal genera were *Mortierella* (7.00%), *Inocybe* (5.36%) and *Neonectria* (3.63%) for soil and *Didymella* (27.30%) *Alternaria* (8.95%) and *Cryptococcus* (4.44%) for leaf (Fig. 3ef). Plant bacterial pathogens (2.29%) were the potential function of the core bacterial taxa in leaves but not in soil; soil had a greater abundance of nitrogen cycle-related function (50.98%) than did leaves (17.55%) (Fig. 3cd, Table S1-2). The abundance of plant fungal pathogens was greater in leaves (68.20%) than in soil (33.93%) (Fig. 3gh, Table S3-4).

We further correlated the top 30 microbial genera of leaf and soil inocula with seed germination and seedling mortality in response to inoculation with nonsterile inocula at G0. The abundances of both the soil and leaf microbial genera were related to the seedling mortality rate (MR) but not to the germination time (GT) or germination rate (GR) (Fig. 3ij). Specifically, the leaf core bacterial genera *Pseudomonas*, *Sphingomonas*, *Massilia*, *Variovorax*, *Aureimonas*, and *Agrobacterium* were positively correlated with MR, but the soil core bacteria, including *Alphaproteobacteria_unclassified*, *Rhodoplanes*, *Vicinamibacteraceae_unclassified*, and *Pedomicrobium*, were negatively correlated with MR (Fig. 3abi). The leaf core fungal genera *Didymella*, *Pleosporales_unclassified*, *Subplenodomus*, and *Bulleromyces* were positively correlated with MR, but the soil core fungal genera *Mortierella*, *Hypocrea*, *Pochonia* and *Volutella* were negatively correlated with MR (Fig. 3efj).

We obtained 192 cultivable fungal isolates from 40 dead seedlings, with an average of 4.825 isolates per dead seedling (Fig. 4a). Based on the ITS genes of the representative strains (Table S5), they were divided into 7 families. The dominant family was Didymellaceae (relative abundance = 66.15%), and the numerically dominant genera were *Allophoma* (50.52%), *Alternaria* (26.04%) and *Epicoccum* (5.73%) (Fig. 4b, Table S5). The seedling-killing effects of these strains on *A. adenophora* exhibited a significant phylogenetic signal (Pagel's $\lambda = 0.82$, $P = 0.0002$). Overall, numerically dominant *Allophoma* (Didymellaceae) and *Alternaria* (Pleosporaceae) had high seedling mortality (54% – 100%) (Fig. 4c, Fig. S3).

Enrichment of microbial community and function by *A. adenophora* seedlings under different treatments

NMDS and PERMANOVA revealed that all four factors significantly affected the bacterial community and functional assembly of the seedlings, and the greatest effects were inoculation time and compartment (all $P < 0.05$, R^2 : 0.102 – 0.138), followed by inoculation source and nutrition (all $P < 0.05$, R^2 : 0.024 – 0.082, Fig. 5a,b,e,f). Additionally, compartment, inoculation source and time significantly affected the fungal community and functional assembly (all $P < 0.05$, R^2 : 0.054 – 0.102), but nutrition affected only the fungal community ($P = 0.001$, $R^2 = 0.031$) (Fig. 5c,d,g,h). Further analysis for each inoculation time treatment showed that compartment and inoculation source mainly affected the microbial community and functional assembly and explained a greater proportion of the variation in bacteria than in fungi and in function than in the community. The nutrient level mainly affected the bacterial community at certain inoculation times ($P < 0.05$ for G0, G21 and G28, Fig. 5i-l, Fig. S4, Table S6-7).

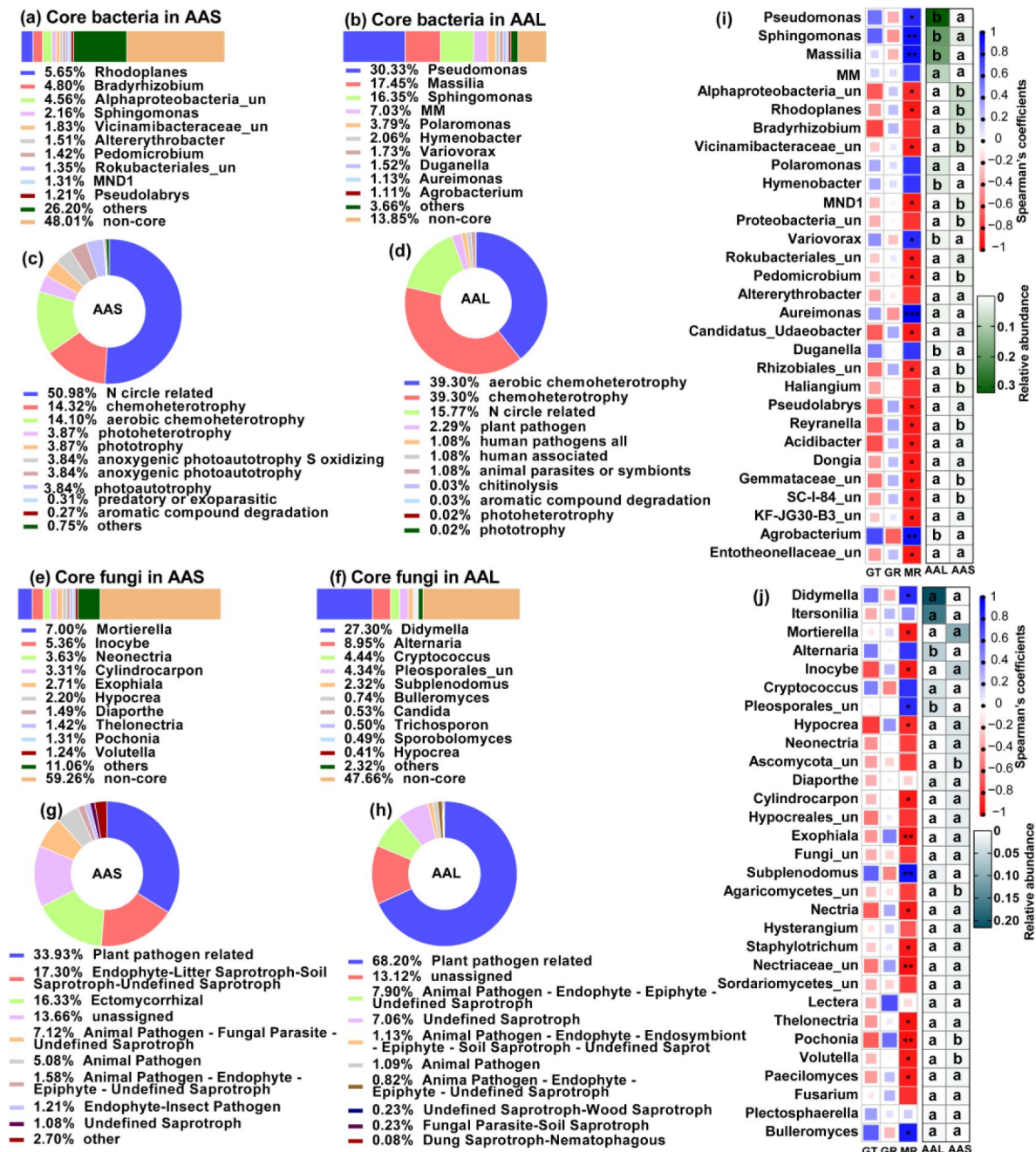


Figure 3.

Microbial community composition and potential functional differences between leaf litter (AAL) and rhizosphere soil (AAS) inocula (a-h), as well as correlations of microbial genera with seed germination and seedling mortality (i-j).

Core bacteria (a-b) and fungi (e-f) in the AAS and AAL groups. The potential functions of the core bacteria (c-d) and fungi (g-h) in the AAS and AAL. Correlations of the relative abundance of the top 30 bacterial (i) and fungal (j) genera with germination time (GT), germination rate (GR) and mortality rate (MR). Only the top 10 core taxa and potential functional groups are shown in the figures. The “un” in the figures is the abbreviation for “unclassified”, and the MM is *Methylobacterium-Methylorubrum*. Several bacterial functions classified as N circles related to the AAL and AAS are shown in Table S1-2. Several fungal guilds classified as plant pathogen-related guilds in the AAL and AAS are shown in Tables S3-4. Red and blue represent negative and positive Spearman's coefficients, respectively. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. Different lowercase letters in the heatmap represent significant differences in the relative abundance of the same genus between AAS and AAL ($P < 0.05$).

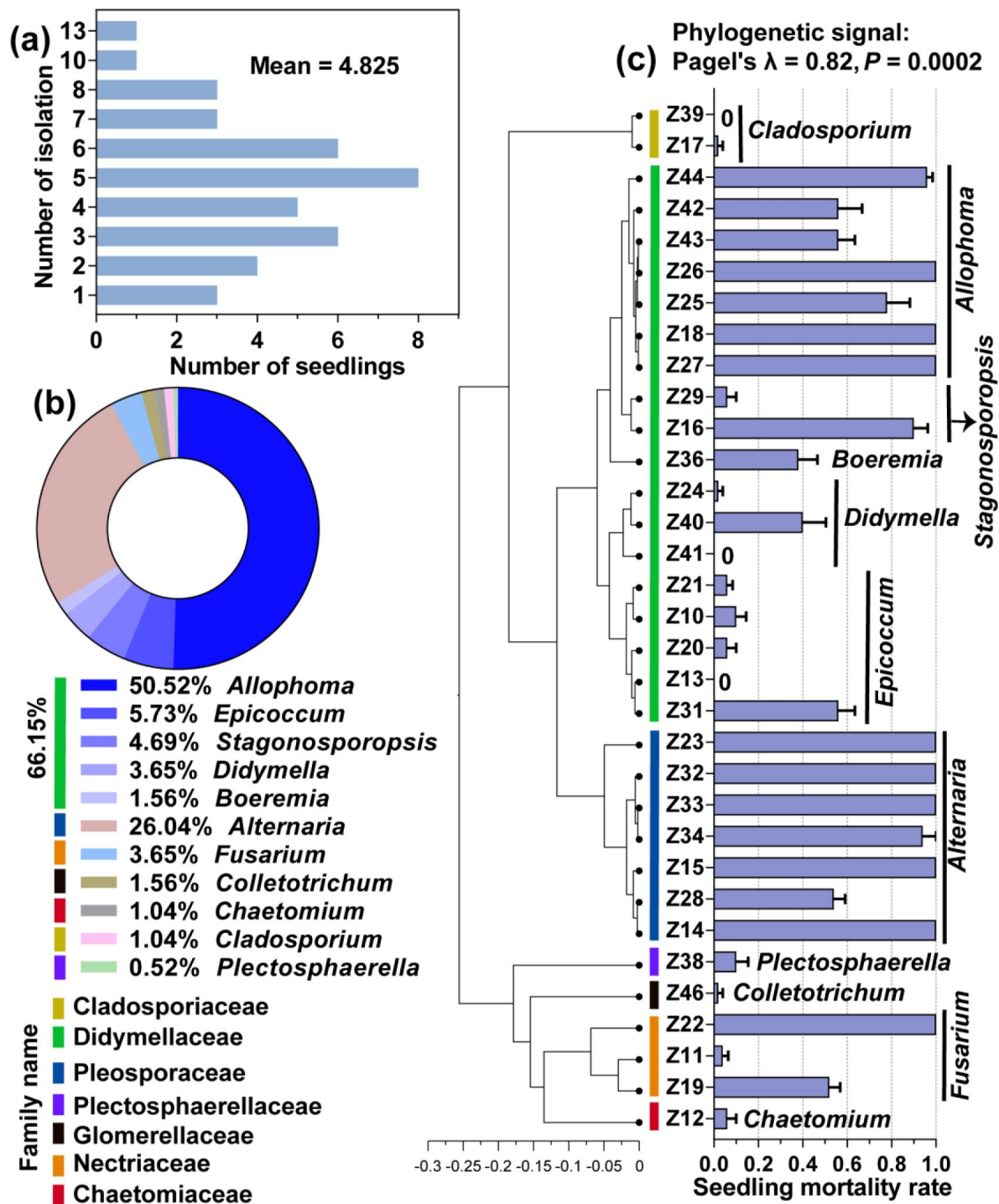


Figure 4.

Cultivable fungi associated with dead *A. adenophora* seedlings and their seedling-killing effects on *A. adenophora*.

Isolation frequency from one dead seedling (a) and cultivable fungal community composition at the genus level (b). The seedling-killing effects of 33 fungal strains on *A. adenophora* and their phylogenetic signals (c).

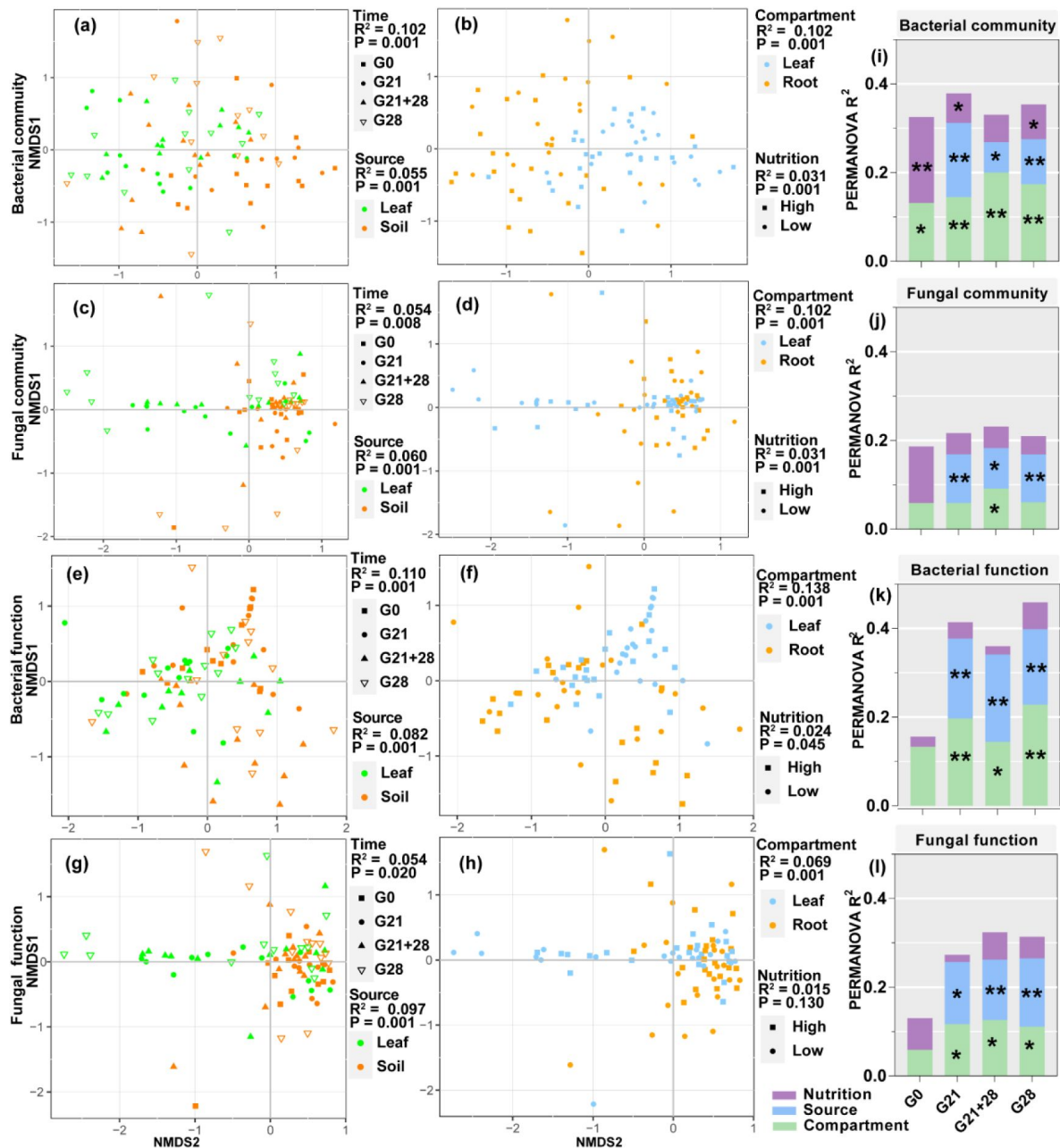


Figure 5.

Enrichment of bacterial and fungal communities and functions by *A. adenophora* seedlings under different treatments.

Nonmetric multidimensional scaling (NMDS) ordinations of Bray-Curtis dissimilarity matrices with permutational analysis of variance (PERMANOVA) of bacterial and fungal communities and function (a-h). Contribution of plant compartment, inoculation source and nutrient level to the variation in bacterial and fungal communities and function at each inoculation time based on PERMANOVA (i-l). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Correlations of the enriched microbial community and function with *A. adenophora* seedling growth

We further analysed the correlation of microbial abundance and putative functions enriched by seedlings with the microbial effect on seedling growth (RI). We identified 47 root endophytic genera that were significantly correlated with *A. adenophora* growth. Among these genera, seven negative genera were less abundant in seedlings treated by leaf inoculation than in those treated by soil inoculation but were similar in abundance in seedlings treated by different inoculation time, and in seedlings grown at different nutrient levels. In contrast, forty positive genera, e.g., the fungi *Duganella* and *Mortierella* and the bacteria *Massilia*, *Pseudomonas*, and *Sphingomonas*, were more abundant in seedlings treated by leaf inoculation than by soil inoculation but less abundant in seedlings inoculated at G0 than at the other three inoculation times (Fig. 6a). Eighteen leaf endophytic genera with significant correlations with the RI were identified, of which three negative genera, namely, the bacteria *Tardiphaga* and *Brevundimonas* and the fungus *Microsphaera*, were more abundant in seedlings inoculated at G0 than at the other three inoculation time treatments and were slightly more abundant in the seedlings inoculated with soil than in those inoculated with leaf; in contrast, fifteen positive genera, e.g., the fungi *Hypocrea* and *Pleosporales_unclassified*, were more abundant in the seedlings inoculated with leaf than in those inoculated with soil and were more abundant in the seedlings inoculated at G21 than in the seedlings inoculated at the other three inoculation time treatments (Fig. 6b).

We identified several bacterial functions in the roots and fungal function guilds enriched in the roots and leaves of *A. adenophora* seedlings that were significantly correlated with the RI (Fig. 6c-e). Two bacterial functions involved in the N cycle (nitrate ammonification and nitrite ammonification) in roots showed a significant positive correlation with RI (Fig. 6c). The fungal guild Ectomycorrhizal showed a significantly positive correlation. Unexpectedly, the putative plant pathogen guild showed a significant positive correlation with seedling growth, while the arbuscular mycorrhizal guild showed a negative correlation (Fig. 6d, e). The positive bacterial functions in roots and fungal guilds in leaves had greater relative abundances in the seedlings after litter inoculation than those after soil inoculation (Fig. 6c, e); additionally, the positive fungal guilds in roots and leaves had significantly greater abundances in the G21 seedlings than in those at the other three inoculation time treatments. Surprisingly, there was no difference in the relative abundance of microbes or functions involved in seedling growth under different nutrient levels (Fig. 6a-e).

Discussion

Leaf litter microbes had more adverse effects on *A. adenophora* seed germination and seedling survival than soil microbes

In support of our first expectation, leaf litter had more adverse effects on seed germination and seedling survival than soil (Fig. 1). Leaf litter has been previously reported to have adverse effects on seedling emergence and population establishment (Abbas et al., 2023; Möhler et al., 2021; Zhang et al., 2022) by reducing light and physical barriers and releasing allelochemicals (Abbas et al., 2023; Gross, 1984; Möhler et al., 2018; Zhang et al., 2019). Our study did not directly test the allelopathic effects of leaf litter. However, leaf litter possibly produces allelochemicals that adversely impact *A. adenophora* seed germination time and seedling survival. We observed that sterile leaf litter inoculation caused longer GTs than sterile soil and the control (nothing inoculated) (Fig. 1a). Interestingly, sterile leaf litter inoculation also caused longer GTs than nonsterile leaf litter inoculation, suggesting that some pathways through which leaf microbes

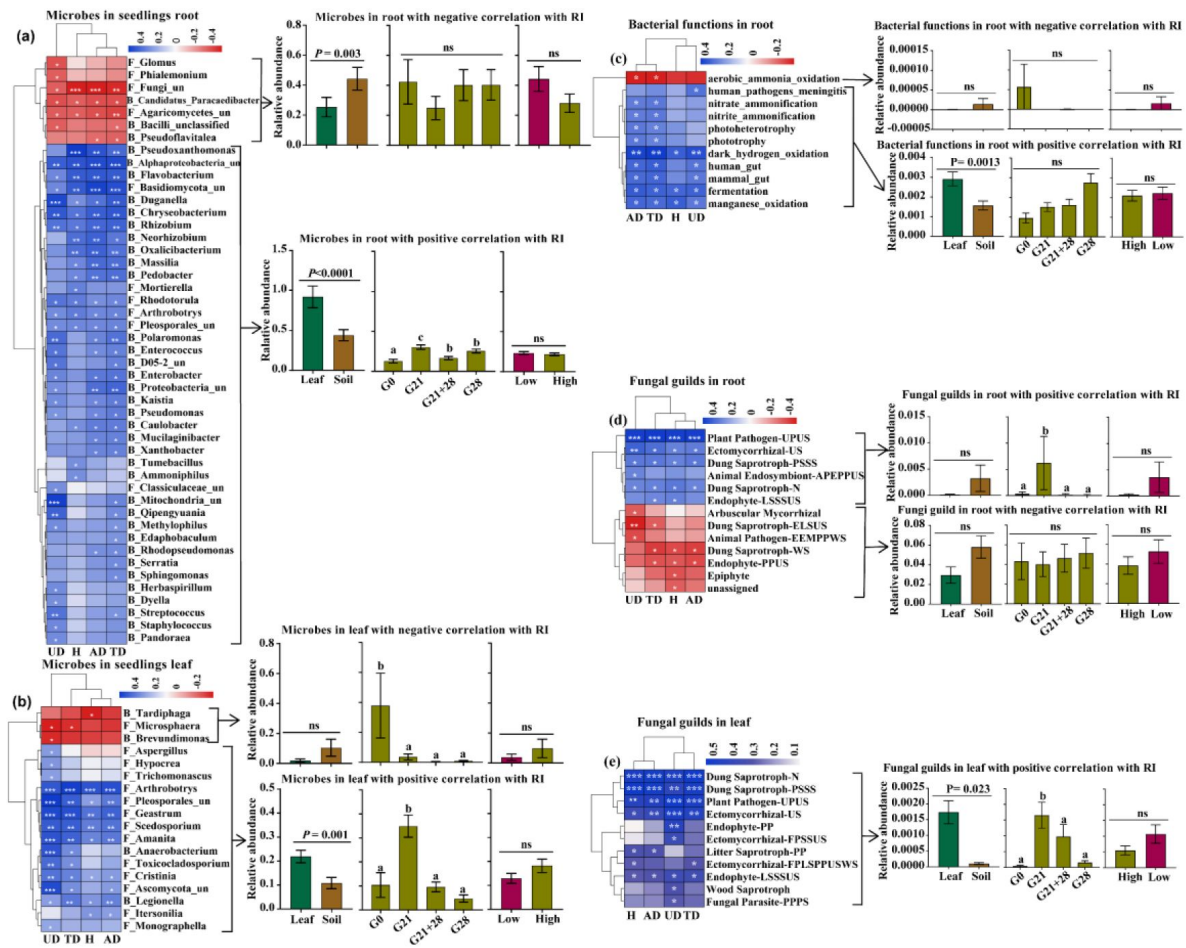


Figure 6.

Correlations of the enriched microbial community and function with *A. adenophora* seedling growth.

These genera accounted for more than 1% of the total relative abundance in seedling roots or leaves. (a) Correlations between 47 out of 214 genera enriched in roots with RI (left) and the relative abundances of genera under different treatments, i.e., different inoculation sources and time and nutrient levels (right). (b) Correlations between 18 out of 184 genera enriched in leaves with RI (left) and the relative abundances of genera under different treatments (right). (c) Correlations between putative bacterial functions enriched in roots with RIs (left) and the relative abundances of functions with negative and positive correlations under different treatments (right). (d) Correlations between fungal guilds enriched in roots with RIs (left) and the relative abundances of guilds with negative and positive correlations under different treatments. (e) Correlations between fungal guilds in leaves with RIs (left) and the relative abundances of guilds with positive correlations under different treatments (right). No bacterial functions in the leaves showed a significant correlation with seedling growth. "F_" represents fungal genera, "B_" represents bacterial genera, "un" represents unclassified; H: RI of seedling height; AD: RI of aboveground dry biomass; UD: RI of underground dry biomass; TD: RI of total dry biomass. Red and blue represent negative and positive Spearman's coefficients, respectively. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. For abbreviations of fungal function guilds, please see Table S8.

alleviate the adverse effects of leaf allelopathy on GTs are unknown. Moreover, sterile leaf inoculation at G0 caused a 19.7% mortality rate for seedlings growing in petri dishes (**Fig. 1c**), but no dead seedlings were observed when the plants were not inoculated (**Fig. 1a**, S1).

Nonetheless, our study highlighted the adverse microbial role of leaf litter in seedling mortality because nonsterile leaves have significantly greater seedling mortality (96.7%) than sterile leaves (19.7%) (**Fig. 1c**). Indeed, we found that leaf litter harbored more abundant bacterial and fungal genera associated with seedling mortality, as well as a greater proportion of plant pathogens, than soil (**Fig. 3**). The results implied that the litter harboured many plant pathogens and thus played an essential role in mediating *A. adenophora* population density by killing conspecific seedlings. These findings provide novel insights for understanding plant invasion. Invasive plants are commonly characterized by rapid growth and a high yield of seeds (Baker, 1974; Parsons WT, 1992; Zheng et al., 2009). These traits are beneficial for rapid population establishment and range expansion at newly invaded sites. However, these invasive plant species commonly form high-density monocultures once the population is established, e.g., *A. adenophora*. In such a situation, high-density seedlings may exacerbate intraspecific competition. Self-limiting of the population elicited by leaf microbes may in turn help *A. adenophora* maintain monocultures at established sites. Thus, it is highly interesting to determine whether leaf microbe-mediated self-limitation at an early life stage is common and important in other invasive systems.

Peripheral microbial sources had more adverse effects on seedling survival and growth when inoculating at the early growth stage than at the later stage

Consistent with our second expectation, inoculation time significantly affected seedling survival and growth; in particular, seedling mortality was greater and seedling growth was poorer when inoculated at G0 than at later growth stages (**Fig. 1c**, d, **Fig. 2b**). The plant growth stage could change the impact of plant host-associated microbes, and such an impact is always strongest in early plant growth stages (Bagchi et al., 2014; Jevon et al., 2020). One potential reason is that small seedlings usually allocate most of their resources to survive and grow, while older seedlings have relatively more resources to defend against pathogen infection (Geisen et al., 2021; Schlöter & Matyssek, 2009). For example, smaller seedlings were more sensitive to inoculated individual fungi, soil microbiota or litter addition than older seedlings due to fewer defense resources and little chance of recovering from biomass loss (Geisen et al., 2021; Jevon et al., 2020; Zhang et al., 2022). Another possible reason is related to the interaction between seed-borne microbes and peripheral microbial sources in young seedlings. There is evidence that seed-borne endophytes are likely to be beneficial for seedling growth and stress resistance (Herrera et al., 2016; Wang & Zhang, 2023). These seed endophytes might be inhibited or even excluded from young seedlings by external sources of microbes inoculated at G0, when seedlings are highly sensitive to inoculated microbes (Geisen et al., 2021; Jevon et al., 2020). Therefore, determining how peripheral microbial sources interact with seed-borne endophytes in seedlings across ontogeny is highly valuable.

We did not observe an adverse effect of leaf litter microbes on *A. adenophora* growth, as observed previously by Kai Fang et al. (2019), who inoculated *A. adenophora* at G0 (sowing seeds), because all seedlings inoculated with leaf litter at G0 died after transplantation into soils in this study. In contrast, both the microbial community and function were significantly positively correlated with seedling growth and had a greater relative abundance in seedlings inoculated with leaf litter than in those inoculated with soil, while those with a negative correlation showed the opposite trend (**Fig. 6**). This finding suggested that leaf litter microbes might have a more positive effect on *A. adenophora* growth than soil microbes if inoculated during the latter growth stage, such as at G21 and G21+28 (**Fig. 2b**). Interestingly, we found that N cycle-related bacterial functions in seedling roots were positively correlated with seedling growth (**Fig. 6c**). Similarly, Zhao et al. (2019) showed that *A. adenophora* invasion increased N cycle-related bacterial functional genes in soil

and subsequently directly promoted plant growth and invasion. K. Fang et al. (2019) [also](#) reported that several root endophytic nitrogen-fixing bacteria of *A. adenophora* could significantly promote its growth. Interestingly, the abundance of these N cycle-related bacterial functions was greater in seedling roots inoculated with leaf litter than in those inoculated with soil (**Fig. 6c**), which suggests a possible way in which some N cycle-related bacteria associated with leaf litter may migrate from leaves into roots after leaf litter inoculation.

Nutrient levels did not affect seedling mortality and microbe-mediated *A. adenophora* growth

Nutrient addition can promote severe invasion, as invasive plants often have greater nutrient availability than do native plants. (Shan et al., 2023 [Slate et al., 2022](#)). However, it is unclear whether such an advantage is involved in a change in the microbial community-driven host growth effect. We also found that *A. adenophora* grew larger and more rapidly in the high-nutrient treatment than in the low-nutrient treatment (Fig. S6); moreover, the nutrient level significantly changed the microbial community and bacterial function (**Fig. 5**). Nonetheless, in contrast to our third expectation, seedling mortality was not affected by seedling growth under different nutrient levels, and nutrient levels negligently affected overall microbe-mediated *A. adenophora* growth and the relative abundance of microbes and functions correlated with seedling growth (**Figs. 1, 2, 6**). Previously, Ai et al. (2018) [reported](#) that nutrient additions cause crops to enrich some bacteria and fungi from soil and increase yield; however, there is no evidence that the increased yield effect is due to enriched microbial communities in this study. Our data indicated that the invasion advantage driven by high nutrient availability may be driven primarily by plant physiological traits, such as rapid nutrient absorption and growth strategies, rather than by enriched microbes. Alternatively, it is possible that our delayed harvest of seedlings under low nutrient levels may cover the distinct microbial role in seedling growth between the two nutrient levels (see Methods).

However, there was an interaction effect between nutrient level and inoculation time on seedling growth (**Fig. 2**). For example, a high nutrient level resulted in a more significant positive microbial effect on seedling growth than a low nutrient level when inoculated at G21, regardless of leaf litter or soil inoculation. It is unclear whether, during the first 21 days before inoculation, more beneficial seed endophytes are enriched to produce a more positive effect on seedling growth under high-nutrition conditions than under low-nutrition conditions, as seed endophytes can facilitate nutrient acquisition and subsequently promote plant growth (Khalaf & Raizada, 2016 [Sanchez-Lopez et al., 2018](#) [Shao et al., 2021](#)).

The same microbial genera had distinct effects on *A. adenophora* seedling survival versus growth

Correlation analysis of the microbial community and function with seedling survival and growth revealed that several genera showed distinct correlations with seedling survival and growth. For example, the bacterial genera *Pseudomonas*, *Sphingomonas* and *Massilia* are positively correlated with seedling mortality and subsequent seedling growth (**Figs. 3, 6**). Many strains belonging to these genera have been reported to promote the growth of many plant species (Jimenez et al., 2020 [Luo et al., 2019](#) [Qiao et al., 2019](#)), including *A. adenophora* (Chen et al., 2019 [K. Fang et al., 2019](#)), because they are commonly involved in N₂ fixation (Albino et al., 2006). However, some microbes, e.g., the fungi *Mortierella* and *Hypocrea*, are negatively correlated with early seedling mortality but positively correlated with later seedling growth (**Figs. 3, 6**). These fungi, as plant growth-promoting fungi (PGPFs), have been widely reported (Contreras-Cornejo et al., 2009 [Ozimek & Hanaka, 2021](#) [Wani et al., 2017](#)). These findings suggested that microbial interactions are highly complicated during the early life stage of *A. adenophora*. On the one hand, there may be sequential effects for some plant growth-promoting microbial groups. For example, the bacteria *Massilia*, *Pseudomonas* and *Sphingomonas* may negatively affect

seedling growth and even kill seedlings if the arrival time is too early after germination. On the other hand, such distinct effects of these bacterial groups on seedling survival versus growth may result from different species from the same genus or even from genetically distinct strains from one species. Interestingly, we found that most *Pseudomonas* and *Sphingomonas* ASVs enriched in the seedlings (>80%) were not associated with the inoculum source (Fig. S8). This suggested that most of the bacterial ASVs positively correlated with growth might be from seeds rather than from the inocula. It is necessary to isolate these enriched microbes to test their interactions with the early life stage of *A. adenophora*.

Surprisingly, related plant pathogen guilds showed a positive correlation with *A. adenophora* seedling growth (Fig. 6). Because these putative plant pathogens were classified as plant pathogens based on the current database, it remains to determine whether such putative plant pathogens for most native plant species are not detrimental to invader *A. adenophora* growth. Indeed, plant pathogens often can switch from a beneficial endophyte to a pathogen or vice versa depending on different host plant species (Delays et al., 2013; Tian et al., 2020).

Seedling-killing microbes were those associated with leaf litter

We found that most seedling-killing microbes isolated from dead seedlings were previously reported to be leaf spot pathogens. For example, *Alternaria* (Pleosporaceae) and several genera belonging to the family Didymellaceae, such as *Allophoma*, *Stagonosporopsis*, *Didymella*, *Boeremia*, and *Epicoccum*, caused high seedling mortality (Fig. 4). *Alternaria* sp. is often pathogenic to a large variety of plants, such as those causing stem cancer, leaf blight or leaf spot (Leiminger et al., 2015; Thomma, 2003; Vergnes et al., 2006), and members of *Allophoma* have also been reported to cause dieback (Babaahmadi et al., 2018) and leaf spot (Garibaldi et al., 2012). All these fungi are leaf spot pathogens of *A. adenophora* and its neighboring native plant (Chen et al., 2022; Kai Fang et al., 2021).

In particular, the numerically dominant *Allophoma* strains obtained in this study had the same ITS genes as the leaf endophyte and leaf spot pathogen *Allophoma* associated with *A. adenophora* (Chen et al., 2022; Kai Fang et al., 2021; Yang et al., 2023). Interestingly, a previous report revealed that the dominant genera in healthy seedlings inoculated with leaf litter were *Didymella* and *Alternaria* (Kai Fang et al., 2019). We did not isolate fungi from healthy seedlings to determine whether the live seedlings indeed lacked or accumulated a lower abundance of the seedling-killing strains than did the dead seedlings in this study. We could assume that these fungal genera likely exist in *A. adenophora* mature individual experiencing a lifestyle switch from endophytic to pathogenic and play an essential role in limiting the population density of *A. adenophora* monocultures by killing seedlings only at very early stages. Thus, it is worth exploring the dynamic abundance of these strains and host resistance variation during *A. adenophora* seedling development.

Implications for developing biocontrol agents for *A. adenophora* invasion

Our data also have implications for the development of biocontrol agents for *A. adenophora* invasion. Currently, several leaf spot fungi, such as the leaf spot fungus *Phaeoramularia* sp., which is released against *A. adenophora* (Kluge, 1991); the white smut fungus *Entyloma ageratinae* against *A. riparia* (Barton et al., 2007); the rust fungus *Uromycladium tepperianum* against the weed *Acacia saligna* (Wood, 2012); and the rust fungus *Puccinia spegazzinii* against *Mikania micrantha* (Day & Ridg, 2019), have been used as biological agents for the control of plant invasion. These agents mainly control weeds by damaging the leaves, stems, and petioles and reducing growth rates, flowering, percentage cover and population density. In this study, the strains associated with leaf litter, such as *Allophoma* sp. and *Alternaria* sp., caused high seedling mortality and thus could control *A. adenophora* invasion at the seedling establishment stage. On the other hand, we found that an external source of microbes had a greater adverse effect on

seedling survival and growth when inoculated at G0 than at the later growth stage. Therefore, prevention and control measures by microbial agents taken at the early seedling stage of invasive plants may be more effective than at the mature stage.

Materials and Methods

Sample collection and preparation

All seeds, rhizosphere soil (AAS) and leaf litter (AAL) of *A. adenophora* were collected from Xishan Forest Park, Kunming city, Yunnan Province (25°55'34"N; 102°38'30"E, 1890 m), on 9th April 2022. We collected dead leaves (litters) from the stems as inoculated leaves to avoid contamination by soil microbes; moreover, dead leaves could better represent litter in natural systems than fresh leaves. All leaf litter and soil samples were collected from five *A. adenophora* populations ~200 m away from each other and treated as independent biological replicates. These *A. adenophora* plants had been grown in monoculture for more than 10 years; thus, their rhizosphere soils and leaf litters were used in our feedback experiment rather than via a typical two-phase approach (the first conditioning phase and the second testing phase) (Brinkman et al., 2010 [\[1\]](#)). The collected soil and litter samples were naturally dried in a clean room and weighed. The soil was ground to a 2 mm sieve before weighing. For convenience in the inoculation application, we prepared these samples in leaf litter bags (each containing 2 g of leaf litter) (Zaret et al., 2021 [\[2\]](#)) and soil bags (each containing 5 g of soil) as well as 0.1 g of soil or litter (cut into pieces smaller than 2×2 mm) in centrifuge tubes. All sample bags and tubes were divided into nonsterile and sterile groups and stored at 4°C until inoculation. The sterile groups (soils or litter) were sterilized by gamma irradiation (30 kGy, 30 h, Huayuan Nuclear Radiation Technology Co., Ltd., Kunming, China), which can kill all microorganisms because no colonies were formed after 7 days of inoculation on PDA media for gamma-irradiated samples (see Fig. S5); moreover, there was no evidence that this irradiation method changed the chemistry of the samples. The nonsterile groups (soils or litter) were natural samples containing live microorganisms. The natural soil or litter (0.3 g) was weighed into tubes and placed at –80°C until DNA extraction.

Experimental design

The experimental design is shown in Fig. 7 [\[3\]](#). (1) We inoculated *A. adenophora* with nonsterile rhizosphere soil and leaf litter of *A. adenophora* and sterile groups as control groups to distinguish the effects of aboveground microbes from those of underground microbes. (2) We inoculated the soils or leaf litters of *A. adenophora* at three growth stages (0 d, 21 d, 28 d) to explore the susceptibility of the seedlings to microbial infection and growth effects. We performed inoculation on the day of sowing the seeds on water agar plates (containing 5 g of agar and 500 mL of water) for germination, which we named G0 inoculation; on the 21st day after seedling growth, we named G21; and on the 28th day after seedling growth, we named G28. Leaf inoculation at G28 was performed to simulate natural microbial spread from the leaf litter to the above part of the seedlings by suspending the leaf bag over the transplanted seedlings without direct contact all the time (see Zaret et al. (2021) [\[2\]](#)). This method may result in only microbial species with easy air transmission to infect seedlings. Thus, an additional combination inoculation (named G21+28) was performed on both the 21st (with seedling contact) and 28th days (without seedling contact) to ensure that most leaf microbes had the opportunity to reach the seedlings. Therefore, we had four treatments for inoculation time. (3) We grew all the inoculated seedlings in background soil (made of Pindstrup substrate, perlite and vermiculite at a volume ratio of 8:1:1, and nutrient content of Pindstrup substrate, see Table S9) by adding the same volume of Hoagland nutrient solution (high nutrient level) or only water (low nutrient level) to the soil. In total, our experimental design included 4 inoculation time (G0, G21, G28, G21+28) × 2 inoculum sources (leaf litter, rhizosphere soil) × 2 microbial treatments (sterile, nonsterile) × 2 nutrient levels (high, low) × 5 replicates = 160 cups. All the cups were randomly placed in the growth chamber and rearranged randomly every week to mitigate potential positional effects. Seedlings were harvested after 8 weeks of growth

under high-nutrient conditions because they grew too fast and touched the PTFE cover; however, we harvested those plants grown under low-nutritional conditions after another 4 weeks of growth due to their very small size (see Fig. S6). No seedlings survived at the G0 inoculation of nonsterile leaf litters when harvested. Stem height, dry aboveground biomass and underground biomass were measured at harvest. Fresh seedling leaves and roots (0.3 g) from three seedlings per treatment as three replicates were harvested and surface-sterilized for microbial community detection. The aboveground and underground dry biomasses for seedlings with less than 0.3 g fresh weight were obtained by linear regression (see Fig. S7). Seed germination and inoculation manipulation for each inoculation treatment are detailed in supplementary method 1 (also see Fig. 7 [↗](#)).

Molecular sequencing of the microbial community

To link microbial sources (leaf litter and soil) with seed germination, seedling mortality and subsequent seedling biomass, we sequenced the microbial community associated with inoculum samples (natural *A. adenophora* soils (AAS) and leaf litter (AAL)), as well as fresh leaves and roots of *A. adenophora* seedlings grown in the nonsterile treatments. Fresh leaves and roots (0.3 g) of each seedling at harvest were chosen for surface sterilization and then stored at -80°C until total DNA was extracted. For DNA extraction, target-gene amplification and sequencing steps, see Supplementary Method 2.

Seedling-killing fungus experiment

No dead seedlings were observed from Petri dishes inoculated with nonsterile soils at G0. Thus, we used 40 dead seedlings obtained from Petri dishes inoculated with nonsterile leaf litter at G0 to isolate fungi. Each dead seedling was cut into $1\text{ mm} \times 1\text{ mm}$ pieces, and three tissues were placed on each PDA Petri dish and incubated at ambient temperature ($20\text{--}25^{\circ}\text{C}$) for 6–8 days or until mycelia grew. Hyphal tip cultures were subsequently transferred onto new PDA plates and incubated until pure colonies appeared. The DNA of these purified fungal strains was extracted and identified by sequencing the ITS region (for details, see Supplementary Method 2).

To test the seedling-killing effects of these fungal strains on *A. adenophora*, sixteen surface-sterilized *A. adenophora* seeds were sown in a water agar plate in a Petri dish. Ten similar-sized seedlings in one Petri dish 21 days after sowing were selected for fungal inoculation. Five Petri dishes were used as five replicates for each strain. Fungi were grown on PDA for 7 days in an incubator at 25°C , after which 3 mm diameter agar discs with fungal mycelia were inoculated into seedlings by touching the leaves or stems (see Fig. S3). Seedlings were regarded as dead when the leaf and stem became brown and rotten. We recorded the number of dead seedlings after 14 days of inoculation with agar discs and then calculated the mortality rate ($\text{MR} = \frac{\text{the number of dead seedlings}}{10}$).

Statistical analysis

It is unreasonable to directly compare seedling biomass among treatments because of different harvest time under high or low soil nutrition conditions (see methods description above), the response index (RI) was calculated to evaluate the feedback intensity (or direction) of microbes in the inocula soil or leaf on seedling growth: $\text{RI} = (\text{variable}_{\text{nonsterile}} - \text{variable}_{\text{sterile}}) / \text{variable}_{\text{sterile}}$ (Bruce Williamson & Richardson, 1988 [↗](#)), a one-sample T test was used to determine the significance between the RI value and zero, where $\text{RI} > 0$ and < 0 represent microbes that promote or inhibit seedling growth, respectively. Because the mortality rates of some sterile groups were zero and their RIs were impossible to calculate, we had to directly compare the seedling mortality caused by nonsterile with by sterile samples and perform the analysis of correlation between the mortality rate and microbial composition. Generalized linear models (GLMs) with Gaussian error distributions (identity link) generated by the “lme4” package were used to identify the effects of inoculation source, time, nutrient level treatments and their interaction on the RIs of plant growth.



Figure 7.

Schematic diagram of soil or litter inoculation at different growth stages.

The abbreviations G0, G21, G28, and G21+28 are shown in [Fig. 1](#).

The R^2 values of the models were obtained by the “piecewiseSEM” package, and P values were estimated using the ANOVA function via chi-squared (χ^2) tests in GLMs. The nonparametric Mann–Whitney U test was used to perform all two-group comparisons, and the Kruskal–Wallis test was performed to compare the differences in mortality rate, RI or microbial relative abundances among the four inoculation time treatments.

Nonparametric Mann–Whitney U tests, Kruskal–Wallis tests and one–sample T tests were performed using SPSS v. 22.0 (SPSS, Inc., Chicago, IL, USA). Nonmetric multidimensional scaling (NMDS) analysis was used to visualize the similarities in bacterial and fungal composition and function among the treatments. Permutational analysis of variance (PERMANOVA) was performed with the ADONIS function in the R (v. 4.2.0) package “vegan” to test the differences in the bacterial and fungal communities and functions among the treatment groups. Bacterial functional profiles were predicted using functional annotation of prokaryotic taxa (FAPROTAX) (Louca et al., 2016 [↗](#)). Fungal functional guilds were inferred using the program FUNGuild, and guild assignments with confidence rankings of “Highly probable” and “Probable” were retained (Nguyen et al., 2016 [↗](#)). The core microbial taxa were primarily selected from the ASVs that appeared (100% prevalence) among all the samples. Spearman’s correlation analysis was used to link microbial communities in inoculation sources (leaf litter and soils) with seed germination and seedling mortality in Petri dishes of the nonsterile G0 treatment, as well as to link the RI of seedling growth with microbial communities or functions in seedling leaves or roots based on all treatments. A correlation was considered significant when $P < 0.05$. Heatmap plotting was performed in R 4.2.0 with the “pheatmap” package. To examine the phylogenetic signal of the seedling-killing of fungal strains on *A. adenophora*, we calculated Pagel’s λ with the R package “phytools”, which measures the distribution of a trait across a phylogeny. A Pagel’s λ closer to 1 indicated a stronger phylogenetic signal (Pagel, 1999 [↗](#)). The remaining figures were visualized in GraphPad Prism v7.0 (GraphPad Software, Inc., San Diego, CA, USA).

Data availability

The raw bacterial and fungal sequence data are archived in GenBank under accession numbers PRJNA1008375 and PRJNA1008403, respectively. Fungal ITS sequences could be obtained from GenBank under accession numbers OR473386–OR473418.

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Author contributions

Zhao-ying Zeng: conceptualization, formal analysis, investigation, methodology, visualization, writing – original draft, writing – review & editing. Jun-rong Huang, Zi-qing Liu, Ai-ling Yang, Yuxuan Li and Yong-lan Wang: investigation. Han-bo Zhang: conceptualization, writing – review & editing, supervision, funding acquisition, project administration.

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Editors

Reviewing Editor

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Meredith Schuman

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Reviewer #1 (Public Review):

Summary:

The work by Zeng *et al.* comprehensively explored the differences in the effects of leaf and soil microbes on the seed germination, seedling survival and seedling growth of an invasive forb, *Ageratina Adenophora*, and found evidence of stronger adverse effects of leaf microbes on *Ageratina* compared with soil microbes. By further DNA sequencing and fungal strain cultivation, the authors were able to identify some of the key microbial guilds that may facilitate such negative and positive feedbacks.

Strengths:

- (1) The theoretic framework is well-established;
- (2) Relating the direction of plant-microbe feedback to certain microbial guild is always hard, but the authors had done a great job in identifying and interpreting such relationships.

Weaknesses:

- (1) Allelopathic effects can't be directly accounted for;
- (2) The fungal strains accumulated in dead seedlings may also accumulate in live seedlings, thus more evidence is needed to validate the claim by the authors that *Allophoma* and *Alternaria* can increase seedling mortality.

<https://doi.org/10.7554/eLife.95502.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The work by Zeng et al. comprehensively explored the differences in the effects of leaf and soil microbes on the seed germination, seedling survival, and seedling growth of an invasive forb, Ageratina adenophora, and found evidence of stronger effects of leaf microbes on Ageratina compared with soil microbes, which were negative for seed germination and seedling survival but positive for seedling growth. By further DNA sequencing and fungal strain cultivation, the authors were able to identify some of the key microbial guilds that may facilitate such negative and positive feedback.

Thank you very much for your assessment.

Strengths:

- (1) The theoretic framework is well-established.*
- (2) Relating the direction of plant-microbe feedback to certain microbial guilds is always hard, but the authors have done a great job of identifying and interpreting such relationships.*

Thank you very much for your assessment.

Weaknesses:

- (1) In the G0 and G21 inoculation experiments, allelopathic effects from leaf litters had not been accounted for, while these two experiments happened to be the ones where negative feedback was detected.*

We did not directly test the allelopathic effects. However, we actually also recorded seed germination time (GT) and rate (GR), as well as the seedling mortality rate (MR) for those treatments inoculated soil and leaf after sowing 28 days (G28 inoculation). It is allowed us to observe possible allelopathic effect by comparing sterile sample with control (nothing inoculated during the first 28 days). In this version, we added the result of GT, GR and MR for nothing inoculated (treated as control) in Figure 1, and described results as: "When inoculated at G0 period, the sterile leaf inoculation significantly delayed germination time more than soil and sterile leaves inoculation and control (nothing inoculated) (Fig. 1a, $P < 0.05$)" (see Line102-104). We have also discussed this point in the resubmitted version as: "Our study did not directly test the allelopathic effects of leaf litter. However, leaf litter possibly produces allelochemicals that adversely impact *A. adenophora* seed germination time and seedling survival. We observed that sterile leaf litter inoculation caused longer GTs than sterile soil and the control (nothing inoculated) (Fig. 1a). Interestingly, sterile leaf litter inoculation also caused longer GTs than nonsterile leaf litter inoculation, suggesting that some pathways through which leaf microbes alleviate the adverse effects of leaf allelopathy on GTs are unknown. Moreover, sterile leaf inoculation at G0 caused a 19.7% mortality rate for seedlings growing in petri dishes (Fig. 1c), but no dead seedlings were observed when the plants were not inoculated (Fig. 1a, S1).

Nonetheless, our study highlighted the adverse microbial role of leaf litter in seedling mortality because nonsterile leaves have significantly greater seedling mortality (96.7%) than sterile leaves (19.7%) (Fig. 1c)” in Line 289-301.

(2) The authors did not compare the fungal strains accumulated in dead seedlings to those accumulated in live seedlings to prove that the live seedlings indeed accumulated lower abundances of the strains that were identified to increase seedling mortality.

Thanks for your concerns. We have not isolated fungi from healthy seedlings to make a comparative study. However, our team work previously found that the seedling-killing *Allophoma* strains obtained in this study had the same ITS genes as the leaf endophyte and leaf spot pathogen *Allophoma* associated with mature *A. adenophora* individual; some seedling-killing *Alternaria* also occur in healthy seedlings inoculated by leaf litter. We thus assumed that these seedling-killing fungi, e.g., *Allophoma* and *Alternaria*, likely exist in *A. adenophora* mature individual by a lifestyle switch from endophytic to pathogenic, and these fungi can kill seedling only at very early life stage of *A. adenophora*.

Thus, we discussed this point as: “In particular, the numerically dominant *Allophoma* strains obtained in this study had the same ITS genes as the leaf endophyte and leaf spot pathogen *Allophoma* associated with *A. adenophora* (Chen et al., 2022; Kai Fang et al., 2021; Yang et al., 2023). Interestingly, a previous report revealed that the dominant genera in healthy seedlings inoculated with leaf litter were *Didymella* and *Alternaria* (Kai Fang et al., 2019). We did not isolate fungi from healthy seedlings to determine whether the live seedlings indeed lacked or accumulated a lower abundance of the seedling-killing strains than did the dead seedlings in this study. We could assume that these fungal genera likely exist in *A. adenophora* mature individual experiencing a lifestyle switch from endophytic to pathogenic and play an essential role in limiting the population density of *A. adenophora* monocultures by killing seedlings only at very early stages. Thus, it is worth exploring the dynamic abundance of these strains and host resistance variation during *A. adenophora* seedling development.” in Line 432-

1.

(3) The data of seed germination and seedling mortality could have been analyzed in the same manner as that of seedling growth, which makes the whole result section more coherent. I don't understand why the authors had not calculated the response index (RI) for germination/mortality rate and conducted analyses on the correlation between these RIs with microbial compositions.

Thanks so much. Response index (RI) was calculated as:

$$\frac{(\text{variablenonsterile} - \text{variablesterile})}{\text{variablesterile}}$$
)). Because mortality rates of some sterile groups were zero values, it is impossible to calculate their RIs. Relatively, only leaf microbes affect seed germination time (GT), leaf and soil microbes did not affect germination rate (GR) (see Fig. 1a,b). Therefore, we preferred to make a direct comparison of the difference between nonsterile and sterile treatments (also see Figure 1d) to assess microbial effect, and we also conducted a correlation by these values with microbial compositions rather than by RIs (see Fig. 3). We emphasized this point in the Materials and Methods in our resubmitted revision as: “Because the mortality rates of some sterile groups were zero and their RIs were impossible to calculate, we had to directly compare the seedling mortality caused by nonsterile with by sterile samples and perform the analysis of correlation between the mortality rate and microbial composition.” in Line 565-568.

(4) The language of the manuscript could be improved to increase clarity.

We have improved language in the resubmitted version.

Reviewer #2 (Public Review):

Summary:

The study provides strong evidence that leaf microbes mediate self-limitation at an early life stage. It highlights the importance of leaf microbes in population establishment and community dynamics.

Thank you very much for your assessment.

The authors conducted three experiments to test their hypothesis, elucidating the effects of leaf and soil microbial communities on the seedling growth of A. adenophora at different stages, screening potential microbial sources associated with seed germination and seedling performance, and identifying the fungus related to seedling mortality. The conclusions are justified by their results. Overall, the paper is wellstructured, providing clear and comprehensive information.

Thank you very much for your assessment.

Reviewing Editor (Recommendations For The Authors):

In addition to the assessments from the reviewers, we have the following comments on your paper:

(1) The experimental design is complicated with regard to the multiple interacting treatments. The statistical analyses show that the interaction terms are important and significant. In this case, it could be more informative to show the detailed results at the sub-level than at the main level in the main text. For example, the main effects of inoculation sources and nutrients shown in Figure 2 are difficult to interpret, because the effects of inoculation sources and nutrients have important dependencies with each other and other factors such as inoculation time as shown in Figure S3. Therefore, Figure S3 is more informative than Figure 2. Please also be cautious that it would be necessary to clarify this context dependence when showing and citing results of the main effect to avoid any possible misunderstanding, such as the case of Figure 2 and S3.

Thanks for your suggestion. We have deleted Figure 2 and placed Figure S3 in the text as Figure 2. And corresponding results have rewritten as “leaf inoculation caused significantly greater seedling mortality than did soil inoculation ($P < 0.001$); the nonsterile sample caused greater seedling mortality than did the sterile sample, especially leaf inoculation during the G0 and G21 periods. Moreover, nonsterile leaf inoculation at earlier stages significantly increased seedling mortality compared with that at later stages (Fig. 1d, $P < 0.05$). However, seedling mortality did not differ between the high- and low-nutrient conditions, regardless of leaf or soil inoculation (Fig. 1d, both $P > 0.05$).” in Line 109-115.

(2) Response index (RI) is already a measure of microbial feedback effect, so that feedback may not be necessary as an explanatory variable in the model with RI as the response variable.

We are sorry that our writing misunderstood you. Here the word “feedback” (e.g., foliage- or soil feedback) does not represent microbial feedback effect, it means leaf or soil inoculation. We have replaced “feedback” by “inoculation source” in the figures and text for better understanding.

(3) Mortality rate is a ratio. It is unclear whether assuming a Gaussian error distribution is fine in your case. It would be important to check the residual distribution and to see whether data transformation (e.g., log) or using other error assumptions (e.g., binomial) is necessary.

Thanks for your suggestion. As you say, it is not appropriate to use generalized linear models (GLMs) with Gaussian error distributions (identity link) to evaluate seedling mortality, because mortality rate is a ratio, which do not meet normality. Thus, we deleted the result of GLM of seedling mortality and directly compared seedling mortality between different microbial treatments, inoculation time, nutrition level and inoculation source by Mann–Whitney U test and Kruskal–Wallis test (see Fig.1 d). All corresponding results have also been rewritten as “leaf inoculation caused significantly greater seedling mortality than did soil inoculation ($P < 0.001$); the nonsterile sample caused greater seedling mortality than did the sterile sample, especially leaf inoculation during the G0 and G21 periods. Moreover, nonsterile leaf inoculation at earlier stages significantly increased seedling mortality compared with that at later stages (Fig. 1d, $P < 0.05$). However, seedling mortality did not differ between the high- and low-nutrient conditions, regardless of leaf or soil inoculation (Fig. 1d, both $P > 0.05$).” in Line 109-115.

(4) Please be consistent about the wording of different treatment names throughout the texts, tables, and figures. For example, “feedback” should only be used for microbial treatment, but not for inoculation source treatment (e.g., Figure 2). We can say there is an effect of microbial feedback only if we compare sterile vs. non-sterile groups, otherwise, there could be other effects, for example, the allelopathic effect pointed out by Reviewer #1. When writing inoculation, please be specific about whether it is for inoculation time or inoculation source (e.g., within multiple statistical tables in the appendix).

Thanks for your good suggestion. We have changed “different feedback” into “different inoculation source” for better understanding our story.

(5) Please clarify which inoculation periods they are for Figures 1d-g.

Thanks for your good suggestion. We have added inoculation periods in Fig.1.

Reviewer #1 (Recommendations For The Authors):

Specific comments:

Lines 12-15: This sentence is too long and complicated, making it unclear what had been done and what had not in previous studies.

Thanks a lot. We have reorganized this sentence as: “However, how the phyllosphere and rhizosphere soil microbes distinctively affect seedling mortality and the growth of invasive plants across ontogeny under varying soil nutrient levels remains unclear.”.

Line 19: is it appropriate to use “enrich” here?

Thanks. We have changed “Microbial inoculation at different growth stages altered the microbial community and functions enriched in seedlings” into “Microbial inoculation at different growth stages altered the microbial community and functions of seedlings”.

Line 24-25: “litter exhibited phylogenetic signals”? not clear what this means.

Thanks. Significant phylogenetic signals represent the seedling-killing effects of fungal strains on *A. adenophora* were related to phylogenetic relatedness of these strains. So, we have changed “fungal strains isolated from dead seedlings inoculated with litter exhibited significant phylogenetic signals to seedling mortality” into “the *A. adenophora* seedling-killing effects of fungal strains isolated from dead seedlings by non-sterile leaf inoculation exhibited significant phylogenetic signals, by which strains of *Allophoma* and *Alternaria* generally caused high seedling mortality.”

| Line 29: using “in turn” in the first sentence seems weird.

We deleted this.

| Lines 32-33: PSFs are usually positive because of?

We have changed “PSFs have positive effects by escaping soil pathogens and recruiting some beneficial microbes” into “PSFs are usually positive because of escaping soil pathogens and recruiting some beneficial microbes”.

| Line 54: why emphasize “a single soil microbe”?

Although the research of Geisen *et al.*, (2021) assessed the effect of each strain of 34 isolates on seed germination and plant growth, Jevon *et al.*, (2020) focused on the soil microbial community on seedling and adult plants survival. Thus, we changed “a single soil microbe” into “soil microbes”.

| Lines 85-86: “tested their mortality to seedlings”? not clear what this means.

We are so sorry that our writing misunderstood you. We have changed “we also isolated the fungi associated with the dead seedlings and tested their mortality to seedlings.” into “we also isolated the fungi associated with the dead seedlings and tested their seedling-killing effects on *A. adenophora*.”.

Results: no statistics and no references for the statistical tables that could support the results were presented in this section.

We have deleted the inappropriate generalized linear models (GLMs) with Gaussian error distributions (identity link) for evaluating seedling mortality, and all corresponding results have also described (see Line 109-115 and Fig. 1d).

| Lines 100-102: this subtitle reads more like a summary of the following results than a title. All subtitles in the Result section have similar issues (i.e. Lines 148-150, 207-209).

Thanks, we subdivided our Results into four sections and we changed these subtitles as: “Effects of leaf litter and rhizosphere soil on the mortality and growth of *A. adenophora* seedlings”, “Correlations of microbial community composition and potential function with seedling mortality at the early stage”, “Enrichment of microbial community and function by *A. adenophora* seedlings under different treatments”, and “Correlations of the enriched microbial community and function with *A. adenophora* seedling growth”.

| Lines 148-206: since there are a lot of results concerning the microbial composition, I suggest focusing on those that could directly explain the positive or negative feedback. The one concerning diversity (e.g. Figure 3 and corresponding texts) does not seem necessary.

Thanks for your suggestion. We have moved figure 3 into the supplementary figures as Figure S2. To focus on core microbes that could directly explain the positive or negative feedback, we reordered Figure 3, where firstly showed the core soil and leaf bacteria, bacterial functions, as well as core soil and leaf fungi, fungal function (Fig3 a-h); and then showed the correlations of top 30 bacterial and fungal genera from soil and leaf with seedling mortality rate (Fig3 i-j).

| Line 180: *is it not common sense that ectomycorrhiza can only be found in soil?*

Yeah, it is. We have deleted this sentence.

| Line 199: *"the seedling mortality of these strains"? not clear what this means,*

We have changed "The seedling mortality of these strains" into "The seedling-killing of these strains on *A. adenophora*".

| Line 291-292: *I don't see how the authors can distinguish between allelopathic and pathogenic effects based on their results.*

We did not directly test the allelopathic effects. However, we actually also recorded seed germination time (GT) and rate (GR), as well as the seedling mortality rate (MR) for those treatments inoculated soil and leaf after sowing 28 days (G28 inoculation). It is allowed us to observe possible allelopathic effect by comparing sterile sample with control (nothing inoculated during the first 28 days). In this version, we added the result of GT, GR and MR for nothing inoculated (treated as control) in Figure 1, and described results as: "When inoculated at G0 period, the sterile leaf inoculation significantly delayed germination time more than soil and sterile leaves inoculation and control (nothing inoculated) (Fig. 1a, $P < 0.05$)" (see Line102-104). We have also discussed this point in the resubmitted version as: "Our study did not directly test the allelopathic effects of leaf litter. However, leaf litter possibly produces allelochemicals that adversely impact *A. adenophora* seed germination time and seedling survival. We observed that sterile leaf litter inoculation caused longer GTs than sterile soil and the control (nothing inoculated) (Fig. 1a). Interestingly, sterile leaf litter inoculation also caused longer GTs than nonsterile leaf litter inoculation, suggesting that some pathways through which leaf microbes alleviate the adverse effects of leaf allelopathy on GTs are unknown. Moreover, sterile leaf inoculation at G0 caused a 19.7% mortality rate for seedlings growing in petri dishes (Fig. 1c), but no dead seedlings were observed when the plants were not inoculated (Fig. 1a, S1).

Nonetheless, our study highlighted the adverse microbial role of leaf litter in seedling mortality because nonsterile leaves have significantly greater seedling mortality (96.7%) than sterile leaves (19.7%) (Fig. 1c)" in Line 289-301.

| Lines 383-414: *Correlations are not necessarily causations. Sometimes a strong correlation may result from higher-order interaction. The authors should be more cautious about the discussion of microbial function in this section.*

Thanks. We deleted all descriptions of adverse effect or beneficial effect on host plant *A. adenophora* growth and cautiously used "negative correlation or positive correlation" to discuss the functions of these enriched microbes by *A. adenophora*. In the last, we also added a sentence to say: "It is necessary to isolate these enriched microbes to test the interactions with the early life stage of *A. adenophora*."

(see Line 411-413).

Lines 489-490: I don't really understand why the authors performed a combination treatment. What did they expect from such a combination?

Thanks. We described our consideration as: “Leaf inoculation at G28 was performed to simulate natural microbial spread from the leaf litter to the above part of the seedlings by suspending the leaf bag over the transplanted seedlings without direct contact all the time (see Zaret et al. (2021)). This method may result in only microbial species with easy air transmission to infect seedlings. Thus, an additional combination inoculation (named G21+28) was performed on both the 21st (with seedling contact) and 28th days (without seedling contact) to ensure that most leaf microbes had the opportunity to reach the seedlings.” see Line 498-505.

Figure 1: why not use "mortality rate" instead of "death rate"?

Thanks. We have changed “death rate” into “mortality rate” in all corresponding figures and text.

Figure 8: This is a very complicated experimental setup. Why did the authors harvest the plants treated with nutrient addition after the 12th day of the experiment and harvest those without nutrient addition after the 16th day? Why the time lag?

Thanks. We explained this as: “Seedlings were harvested after 8 weeks of growth under high-nutrient conditions because they grew too fast and touched the PTFE cover; however, we harvested those plants grown under low-nutritional conditions after another 4 weeks of growth due to their very small size (see Fig. S6).”

(see Method in Line 514-517).

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